4.1 Writing Equations in Slope-Intercept Form

Name _____ Period ____ Date _____

Algebra I 4.1 · Student sheet

I can…

- write an equation of a line given its slope and y-intercept or its graph.
- write an equation of a line through two points.
- write linear functions to model real-life situations.

Vocabulary · fill in the missing word

Slope-intercept form — y = mx + b, where m is the slope and b is the _____

Linear model — a linear function that _____ a real-life situation

Part 1 · Phone plan

$25 per month plus $50 for the phone.

Months	0	1	2	3	4
Total cost					

Slope: _____ (means: _____) y-intercept: _____ (means: _____)

C(m) = _____

Part 2 · Notes and examples (fill in each note, then work the example)

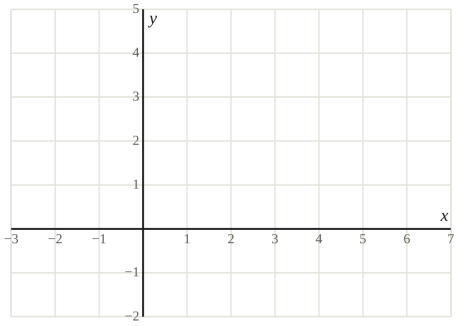
Slope-intercept form: y = mx + b: m is the _____; b is the _____.

Example 1

Write an equation of the line with slope $\frac{3}{4}$ and y-intercept −2.

Example 2

Write an equation of the line shown.



From two points: From two points: find _____ first, then substitute a point to find _____.

Example 3

Write an equation of the line through (2, 1) and (5, 10).

Example 4

A pool has 200 gallons of water and is filled at 50 gallons per minute. Write a function for the number of gallons g after m minutes.

How much water is in the pool after 6 minutes?

Part 3 · Partner practice

Partner 1

Write an equation of the line through (−1, 6) and (3, −2).

Partner 2

f is linear, f(0) = 7, and f(2) = 13. Write f(x).